Internal use only

Space Structure Integrity Battlecard

Summary/Elevator pitch:
Hexagon solution allows all actors of the space and defense industry to ensure the structure they design will withstand the different dynamics loading encountered during launch. What differentiates us is our ability to make a complete assessment based on a single structural model solved with the industry standard MSC Nastran solver.

May also use / Cross sell

- Primary products involved: MSC Nastran, Actran, CAEfatigue
- Space structures are also subjected to other excitation / load cases which may require:
  - Adams – Solar panel deployment, articulated structures, separation systems
  - Dytran – Advanced explicit nonlinear analysis for extreme shock/crash cases
  - Marc & Cradle – Aerodynamic and thermal loads can be accurately predicted and applied to the structure
  - Digimat – Nearly all structures are composite materials which can benefit from multiscale modeling capabilities

Target industries and applications

- Satellite / Payload, Launch vehicles, Rocket Engine, Missiles, Embedded equipment (electronic box, battery, controls…), Space agencies, Research institutes

Key customer wins / References

- Airbus Defense and Space, Thales Alenia Space, Avio, Raytheon, L3 Harris…

Benefits and USPs

- **One-stop shop for space structure design –** Our wide range of capabilities can cover all aspects required for the design of space structures. All capabilities are available through a single MSC One license (reduce cost)
- **Productivity –** Key pre-processing, post-processing and HPC capabilities are allowing for faster turnaround time of results. Allowing to reduce development cycles or perform more iterations
- **Advanced capabilities –** Hexagon has state of the art capabilities in all key areas for space structure design: Structure vibrations, random loads, vibro-acoustics, shock response, fatigue
- **All-in-one model –** Thanks to technology integration, all capabilities are available based on a single MSC Nastran model. This limits the need for error prone model translations and reduces the number of CAE tools required
- **Confidence –** Advanced robust design capabilities are providing confidence that simulation results will be representative of real hardware behavior
- **Expertise/Experience –** With 60+ years of software simulation experience, Hexagon has expertise to support customers through the complete analysis cycle

Pains / Frustrations	Discovery Question	Solution / Product positioning
"We need to exchange and translate models between different tools to assess different loading conditions. This is error prone and time consuming"	• How many software tool is your team using to design for all loading conditions? • How does this translate in terms of license? How many license contracts do you need to maintain?	• We have integrated many key capabilities in MSC Nastran to leverage a single model for all excitation types • MSC Nastran Embedded Vibration Fatigue is our integrated solution for random loads and vibration fatigue. This is typically done with third party tools like Maya or DesignLife • MSC Nastran now integrates diffuse sound field excitation and advanced vibro-acoustic capabilities of Actran. This is typically done with VAOne or Wave6 • MSC Nastran integrates Virtual SEA for shock analysis and high frequency vibration response. This is typically done with VAOne or Wave6
"Our simulation models are increasingly refined as we need to accurately represent the model to have confidence in results. This causes run times to become longer and exchanging large amount of data"	• How would you benefit from running your simulation models faster? • Do you have bottlenecks due to model run time or handling large amount of simulation data?	• MSC Nastran, CAE Fatigue, and Actran have best in class capabilities when it comes to solver performance and solving large models (handling large assemblies with super-elements and modules, Solver and HPC capabilities…) • Integrated solutions for random, acoustic and fatigue are reducing the need to exchange data (smaller results files, no intermediate results files)
"Our engineers need to learn multiple tools and be proficient in multiple areas to deliver results. This requires a lot of time investment for new team members"	• How good is the support from your current vendors? • With how many different support systems do you have to interact to get all the answers you need?	• Complex workflows are simplified and require fewer CAE tools due to the innovative integrated processes based on the original MSC Nastran model. • Our global customer support team can handle any question related to the different type of analysis performed by your team • Our multidisciplinary services team provides training, consulting, and staff augmentation services to bring users up to speed quickly and establish robust and performant processes that can be leveraged effectively by your team.

Space Structure Integrity competitor battlecard

MSC Nastran perception in the industry:
Our competitors will attempt to portray MSC Nastran as an old code which did not get much investments for the past decade. In fact, we made massive improvements in the code and MSC Nastran is not equivalent to the other Nastran codes and still better than other competitors such as Ansys or Abaqus for space structure integrity applications. The Hexagon acquisition is also a good opportunity to stress the fact that we are making new investments and want to double down on customer satisfaction.

To beat the competition, focus on:
- No competitors can provide a complete solution covering all the aspects we can cover. Customers will necessarily be in a situation where they must leverage different tools from different vendors to perform their work.
- With MSC One, customers get access to all our capabilities in a single cost-effective licensing environment. All needs from basic to advanced are covered.
- MSC Nastran is the industry standard for structural analysis (in particular for dynamics) and has benefited from significant improvements in the past years.
- Hexagon D&E strategy consist in developing best in class capabilities in dedicated software (Actran, CAEfatigue) and integrate the most relevant capabilities to MSC Nastran.

Competitor	Competitor strengths and limitations	Objection handling	Key differentiators
Siemens (Femap, NX Nastran, Maya)	• FEMAP + NX is often seeing as “good enough” compared to MSC Nastran. Good capabilities for less cost. • Recently focused on test + simulation bundling for DFAT (direct field acoustic testing) application	• NX Nastran is only “good enough” because of the cost, not the capabilities. • This cost issue only holds true if comparing MSC Nastran vs NX Nastran but not including the vibro-acoustic and fatigue capabilities that customers would have to include otherwise	• With Apex we have now a more modern pre-processing environment than FEMAP • MSC Nastran is full featured and more performant than NX Nastran • Siemens does not have good vibro-acoustic capabilities for the space industry
Dassault (Abaqus, FESafe, Wave6)	• Abaqus is very complete and able to handle many loadcases application cases • State of the art capabilities for vibro-acoustics with Wave6	• Abaqus is not widely used for structural dynamics as it is not based on Nastran architecture commonly used in Space industry (useful for exchanging information between suppliers) • Wave6 price is very high ($50k for single user, $100k for single node locked license)	• MSCOne token system gives access to all state-of-the-art solvers of Hexagon in a more affordable way
Altair (Hyperworks, Optistruct, SEAM)	• Very strong position in pre/post-processing (Hypermesh used for pre/post of MSC/NX Nastran) • Trying to push Optistruct as a solver based on Hypermesh usage	• Focus on solver capability gaps: ○ Optistruct has important capability gaps compared to the industry standard MSC Nastran. ○ SEAM has very limited vibro-acoustic capabilities compared to Actran	• Our solvers are state of the art when it comes to dynamics, acoustics and fatigue • Apex can show some advantages but only push when clear gaps are identified
Ansys (Mechanical, DesignLife)	• Very well established in various industries + academia. Often the first solution people will consider based on prior knowledge • Very complete offering for space going way beyond structural simulation can lead to enterprise deals	• Ansys is not as versatile as Nastran for dynamic analysis in the space industry • Ansys does not have comparable vibro-acoustics and fatigue capabilities • For fatigue, they must rely on Design Life (by HBK)	• All-in-one package (solution+support) catering to all the needs of the space industry
ESI (VAOne)	• Reference in vibro-acoustics and SEA is particular	• ESI has very limited structural analysis capabilities. Only VAOne is relevant in the space industry (no bundling possible) • VAOne is rapidly losing traction in the market due to inability to innovate, old interface, etc • Cost of VAOne is typically quite high and some companies use this as a driver to change	• Our vibro-acoustic capabilities in Actran (and integrated in MSC Nastran) are more performant and versatile • Better performance on large models • MSCOne for accessing Actran on top of MSC Nastran